TECH HORIZON 2.0

Hackathon Proposal

Edge-AI Based Distributed Fleet Coordination for Autonomous Mobile Robots

Decentralized Intelligence for Faster, Safer and More Resilient Smart Warehouses

Tech Horizon 2.0 Team
Demo-ready for Tech Horizon 2.0 Hyderabad

TECH HORIZON 2.0

Hackathon Proposal

Edge-AI Based Distributed Fleet Coordination for Autonomous Mobile Robots

Decentralized Intelligence for Faster, Safer and More Resilient Smart Warehouses

Tech Horizon 2.0 Team Demo-ready for Tech Horizon 2.0 Hyderabad

1. Executive Summary

Modern warehouse automation increasingly depends on fleets of Autonomous Mobile Robots (AMRs) to move products between storage, pickup and drop-off locations. A conventional centralized fleet architecture places coordination and decision-making at a central controller. As fleet size and operational complexity increase, this model can introduce decision bottlenecks, network dependency, latency, congestion at shared spaces, and a single point of failure.

This proposal presents an Edge-AI based distributed fleet coordination architecture in which each AMR can perceive its environment, make local decisions, communicate directly with peer robots, negotiate conflicts, dynamically re-plan, and continue operating when central connectivity is disrupted.

The proposed hackathon prototype will demonstrate a coordinated fleet of 3–5 AMRs operating in a warehouse environment with narrow aisles, overlapping tasks, dynamic obstacles and simulated network disruption. The target is zero inter-robot collisions, successful conflict resolution and dynamic re-routing, continued operation during network disruption, and at least a 20% reduction in total task completion time compared with a basic stop-and-wait strategy, subject to validation during the demo.

2. Problem Statement

Centralized warehouse robotics systems can become less effective as the number of robots and the complexity of the environment increase. When multiple robots encounter the same constrained area, they may have to wait for a central controller to resolve the conflict. Network latency or Wi-Fi dead zones can further delay decisions, while loss of the central service can pause fleet operations.

- Centralized bottleneck: a single decision point can limit responsiveness as fleet size grows.
- Network dependency: communication delays and connectivity gaps can affect real-time coordination.
- Single point of failure: central-controller failure can interrupt fleet-level operation.
- Choke-point congestion: robots may stop and wait when their paths overlap in narrow aisles.
- Limited local adaptability: task allocation and routing may not respond quickly to local conditions.

Representative scenario: two robots reach a narrow aisle simultaneously. Instead of independently resolving the conflict, both may wait for a central command, creating congestion and avoidable lost time.

3. Proposed Solution

The proposed system distributes intelligence across the fleet. Each AMR follows a local decision loop:

PERCEIVE → DECIDE → COMMUNICATE → COORDINATE → ACT

4. System Workflow

- Perception: LiDAR, cameras and IMU sensors observe the environment and robot state.
- Edge AI processing: lightweight detection and short-term prediction are performed locally.
- Local decision-making: the robot plans its path and performs local collision avoidance.
- Peer coordination: robots exchange position, velocity, task, path, intended direction, obstacle and priority information.
- Conflict resolution: overlapping paths are detected and robots compare priority or estimated arrival time.

- Negotiation: robots adjust speed or re-plan their routes to resolve conflicts.
- Action and confirmation: the robots execute the agreed plan and continue operation.
- Dynamic response: newly detected obstacles trigger local broadcasts and fleet re-planning.

Example: Robot A detects a blocked aisle, broadcasts the obstacle information, Robot B receives the update, and both robots re-plan locally so that the fleet can continue without waiting for a central command.

5. Distributed Coordination Mechanism

Robots periodically broadcast relevant state and intent information to nearby peers, including:

- Position and velocity
- Current task
- Planned path
- Intended direction
- Obstacle information
- Priority level

For a detected conflict, the local coordination process is:

- Detect potential path overlap.
- Compare priority and estimated time of arrival.
- Negotiate by adjusting speed or selecting an alternative route.
- Confirm the resulting plan and continue execution.

The central design principle is distributed decision-making with explicit conflict resolution and deadlock prevention, rather than a simple stop-and-wait behavior.

6. Technical Architecture

6.1 Edge Intelligence

- Perception: lightweight object detection using YOLO variants, semantic cues and LiDAR-based obstacle sensing.
- Localization: SLAM combined with IMU and odometry fusion for robust pose estimation.
- Path planning: A* / D* Lite with dynamic re-planning and velocity profiling for safety.
- Multi-agent coordination: priority negotiation, distributed task allocation and conflict detection.
- Edge targets: NVIDIA Jetson-class hardware, industrial ARM boards or equivalent platforms, with models optimized for latency and power.

6.2 System Layers

Candidate communication and infrastructure technologies:

- Protocols: ROS2 / DDS, WebRTC DataChannel, gRPC
- Networks: Wi-Fi 6 / 5G / TSN where available

7. Why Distributed Edge-AI?

Priority outcomes: LOWER LATENCY • HIGHER RESILIENCE • BETTER FLEET UTILIZATION

8. Hackathon Demonstration Plan

The prototype will use a controlled warehouse simulation or physical AMR setup representing pickup and drop zones connected by aisles. The demonstration will focus on visible, measurable coordination behavior.

- Deploy 3–5 AMRs in a warehouse environment.
- Assign overlapping pickup/drop tasks to create competing route requirements.
- Introduce a narrow aisle or choke point where robot paths can conflict.
- Allow robots to detect the conflict, exchange state and negotiate priority.
- Introduce a dynamic obstacle that blocks an aisle and observe peer-to-peer broadcast and re-routing.
- Simulate network disruption and verify that local robot coordination continues without central approval.
- Record task completion time, waiting time, collision status and continuity during the disruption.

9. Success Criteria and Validation

10. Expected Impact

The proposed architecture is intended to improve warehouse fleet responsiveness and resilience by moving time-critical coordination closer to the robots. The expected operational benefits are:

- Real-time coordination between robots
- Reduced congestion and waiting at shared spaces
- Improved fleet utilization
- Lower dependence on a continuously available central controller
- Resilient operation during network disruption

11. Future Roadmap

The concept can be extended beyond the hackathon prototype through:

- Federated learning for distributed model improvement
- Digital twin integration for warehouse simulation, testing and optimization
- Multi-warehouse coordination
- Human-robot collaboration

12. Conclusion

This project proposes a shift from instruction-dependent warehouse robots toward a fleet of autonomous agents that can perceive, negotiate, adapt and act locally. By combining Edge AI, peer-to-peer communication, distributed conflict resolution and dynamic re-planning, the system aims to reduce latency, improve resilience and make more effective use of the available robot fleet.

Distributed by design. Intelligent at the edge. Autonomous as a fleet.

Capability	Proposed Approach
Local AI inference	Perception and short-horizon planning run on-board using Jetson/CPU-class edge hardware.
Peer negotiation	Robots share position, intent and other coordination information directly with peers.
Dynamic re-planning	Robots locally re-route and reallocate tasks without requiring central approval.
Optional central layer	A central server can remain available for monitoring, analytics, global mapping and high-level management.

Layer	Responsibilities	Resilience / Role
Device Layer	Edge inference, local planning and peer-to-peer low-latency communication.	Continues operating if the server becomes unavailable.
Edge Server Layer	Global map, model management and monitoring; optional central task allocation and analytics.	Supports fleet-level management without being required for every local decision.
Optional Cloud	Long-term analytics, dashboards and OTA updates.	Provides extended management and data capabilities.

Dimension	Centralized Approach	Distributed Edge-AI
Decision-making	Central bottleneck; single point of failure	Local decisions; no single decision bottleneck
Latency	Higher dependency on network; greater latency	Low-latency local response with peer coordination
Resilience	Operations can be affected during outages	Continues operating during network disruption
Scalability	Limited local autonomy as fleet complexity grows	Fleet can scale through peer negotiation

Metric	Target / Validation
Inter-robot safety	Zero inter-robot collisions during the demonstration.
Conflict resolution	Successful local resolution of overlapping paths.
Dynamic re-routing	Successful fleet re-routing around a newly introduced obstacle.
Network resilience	Continued operation during simulated network disruption.
Efficiency	Target ≥20% reduction in total task completion time versus a basic stop-and-wait strategy; expected and to be validated experimentally.